FILOX & FILKGC: LLM-DRIVEN ONTOLOGY EXPANSION AND KNOWLEDGE GRAPH CONSTRUCTION FOR INDIAN LEGAL TEXTS

By Harsh Yadav

Supervisors:

Prof. Vasudha Bhatnagar,

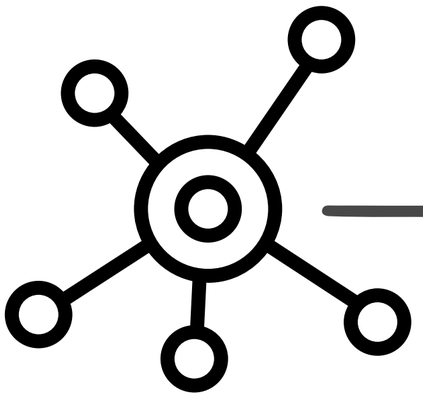
Dr. Vikas Kumar

OVERVIEW

- Introduction..... 2
- Motivation..... 4
- Objective..... 5
- Evolution of Information Extraction (IE)..... 5
- Current Techniques.....12
- Methodology & Pipeline.....14
- Implementation & Experimental Results..... 21
- Conclusion and Future Work..... 30
- References..... 31

INTRODUCTION

- **Indian Legal Documents** - Highly Unstructured
- Need to **Structure** them
- Process:
  - **Named Entity Recognition** (NER), (Rule-based, ML-based) for Entity Extraction
  - **Relation Extraction**
  - **Triplet** Formation (Head, Relation, Tail)
  - Store in **Knowledge Graphs** (KGs)



INTRODUCTION

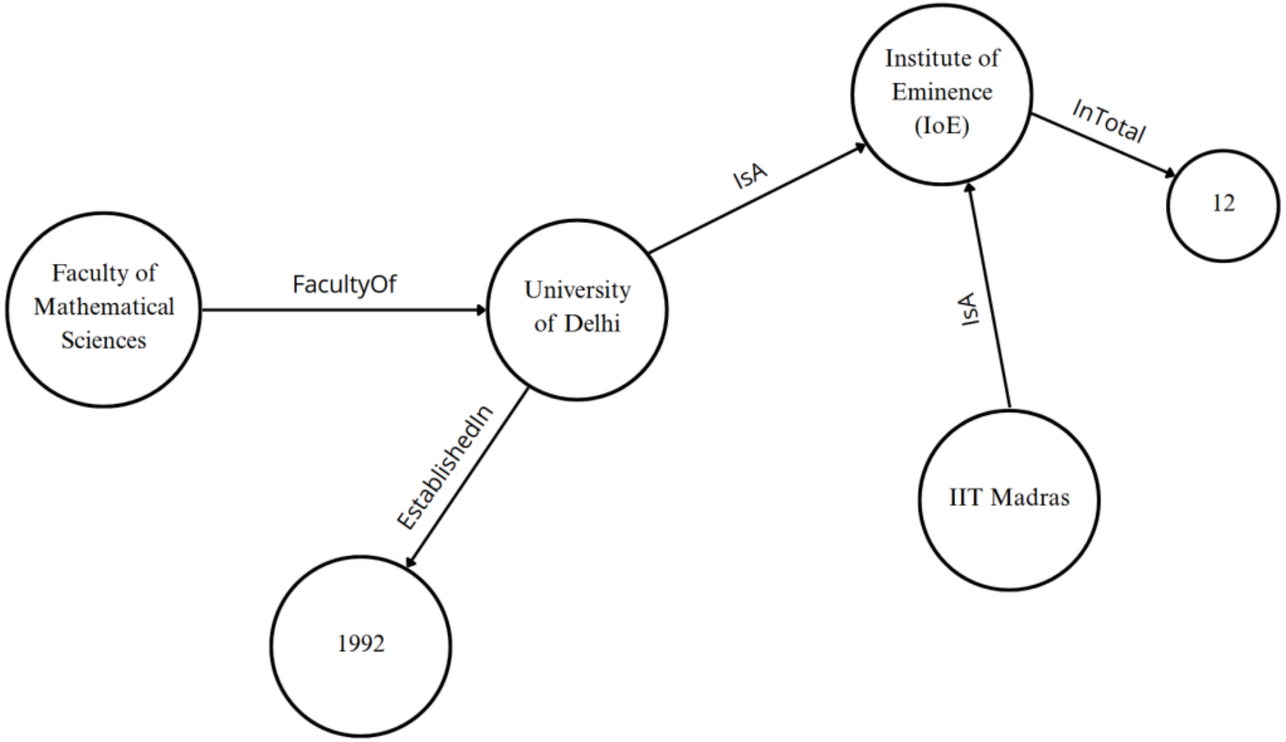


Figure 1.1: An example of entities and relations in a KG

MOTIVATION

- With the help of **high-quality KGs,** following **downstream tasks** can be aided:
  - Document Summarization
  - Semantic Search
  - Legal Chat-Bots
  - Finetuning LLMs
  - Legal Decision Support
  - Precedent Recommendations
  - Risk Assessment
  - and many more....

OBJECTIVE

Firstly, to develop a **framework** that can **expand** the **existing legal ontology** by using **LLMs** in a data-driven manner.

Secondly, to develop an LLM-based **framework** that can **utilize** any such predefined ontology to extract **high-quality triplets** from legal documents to use them for creating **legal knowledge graphs**.

EVOLUTION OF INFORMATION EXTRACTION

- **Traditional Approaches & Challenges**
  - Labour-Intensive Manual Processing
    - time-Consuming
    - manually reading & extracting information
    - high cognitive load
    - prone to human error
    - limited scalability

EVOLUTION OF INFORMATION EXTRACTION

- **Early Computational Approaches & Challenges**
  - **Open Information Extraction** (OIE)
    - introduced by **TextRunner**[1]
    - developed at University of Washington
    - high Ambiguity
    - unable to capture domain-specific nuances
  - Methods introduced later - Reverb[2], OLLIE[3], ClausIE[4], CSD[5], etc.

EVOLUTION OF INFORMATION EXTRACTION

- **The rise of Large Language Models (LLMs)**
  - Made the job easy.
  - **NO** need to **train** every time while adding **new entity categories**, hence, **NO** need of **labelled data** & **huge compute**.
  - Already **pre-trained** on huge data corpus. WIDE CONTEXT
  - **New pre-defined relations** can also be added **easily**.
  - **Few-Shot** prompting and **Persona-Based** prompting techniques helps it further.

EVOLUTION OF INFORMATION EXTRACTION

- Can we **directly inference LLMs** for triplet generation?
  - **Yes**, we can. But, they **won't** be of **high-quality** and would be very **ambiguious**. Reasons:
    - Could often get confused among **Homonyms**.
      - E.g: ✅Legal Context: "The plaintiff filed a **suit** against the defendant for breach of contract." ***(Lawsuit)***
      - ❌Fashion Context: "He wore a formal **suit** to the wedding." ***(Clothing)***

EVOLUTION OF INFORMATION EXTRACTION

- Need to **guide** in labelling **entities** with valid **entity-types** and valid assignment of **relations** among two entities, specific to legal domain, otherwise can produce **hallucinated triplets.**
  - E.g: Sentence: "John Doe filed a lawsuit against ABC Corp for breach of contract in the Supreme Court."
  - ❌Hallucinated Relations: (John Doe, JUDGE_OF, Supreme Court)

EVOLUTION OF INFORMATION EXTRACTION

- Need **canonicalization** of relation terms to **reduce redundancy**, because of the **synonyms** produced.
  - E.g: Sentence: "The court ruled that ABC Corp breached the contract with John Doe."
  - ❌ Non-Canonicalized Variants (Synonyms):
    - (ABC Corp, VIOLATED, contract) ❌
    - (ABC Corp, FAILED_TO_HONOR, contract) ❌
    - (ABC Corp, DID_NOT_UPHOLD, contract) ❌
  - ✅ Canonicalized Relation: (ABC Corp, BREACHED, contract)

CURRENT TECHNIQUES

- Current IE Approaches:
  a. SpaCy's **"NER"**[6] **isn't domain specific**, and extracts **17 entity categories**, and don't extract relations.
  b. "**E-D-C**"[7] **isn't domain specific**. However, it uses LLMs for IE and Triplet Construction. It **does not assign** any category to the entities.
  c. OpenNyai's **"LegalNER"**[8] model is **domain-specific**. However, it extracts only **14 entity categories**, and don't extract relations.
  d. "**Constructing a KG from Indian Text Corpus**"[9] is also **domain-specific**. But, it also extracts only **16 entity categories** and **13 relations.**

CURRENT TECHNIQUES

- Challenges with current approaches:
  - Not **Domain** Specific
  - if **yes**, then **less** entity categories and relation terms, can't cover maximum sub-domains of legal domain.
  - to **add** new categories, need to **train** the existing models each time. NEED **Labelled** Data, **Compute** Hungry !!

METHODOLOGY

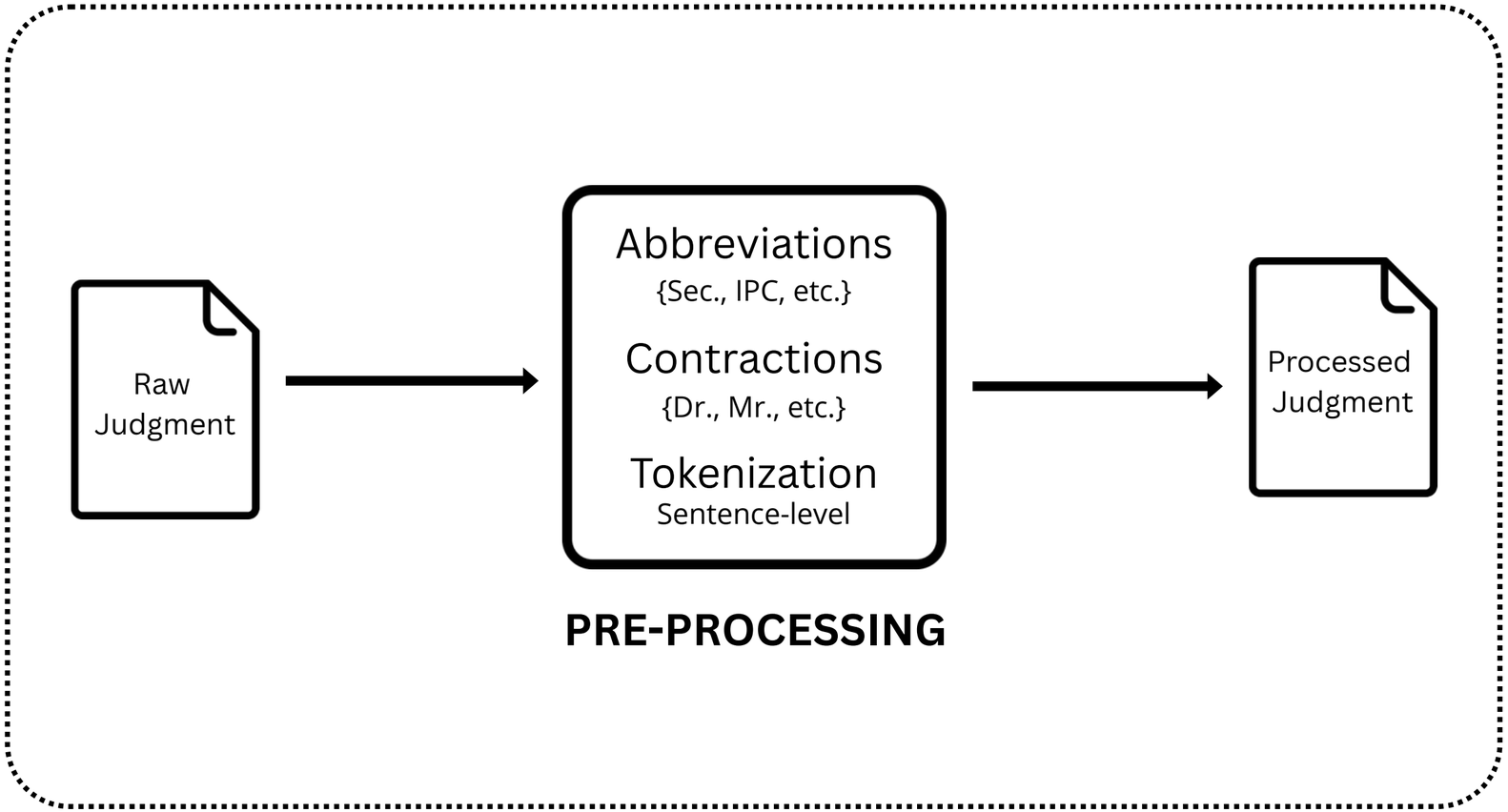
&

PIPELINE

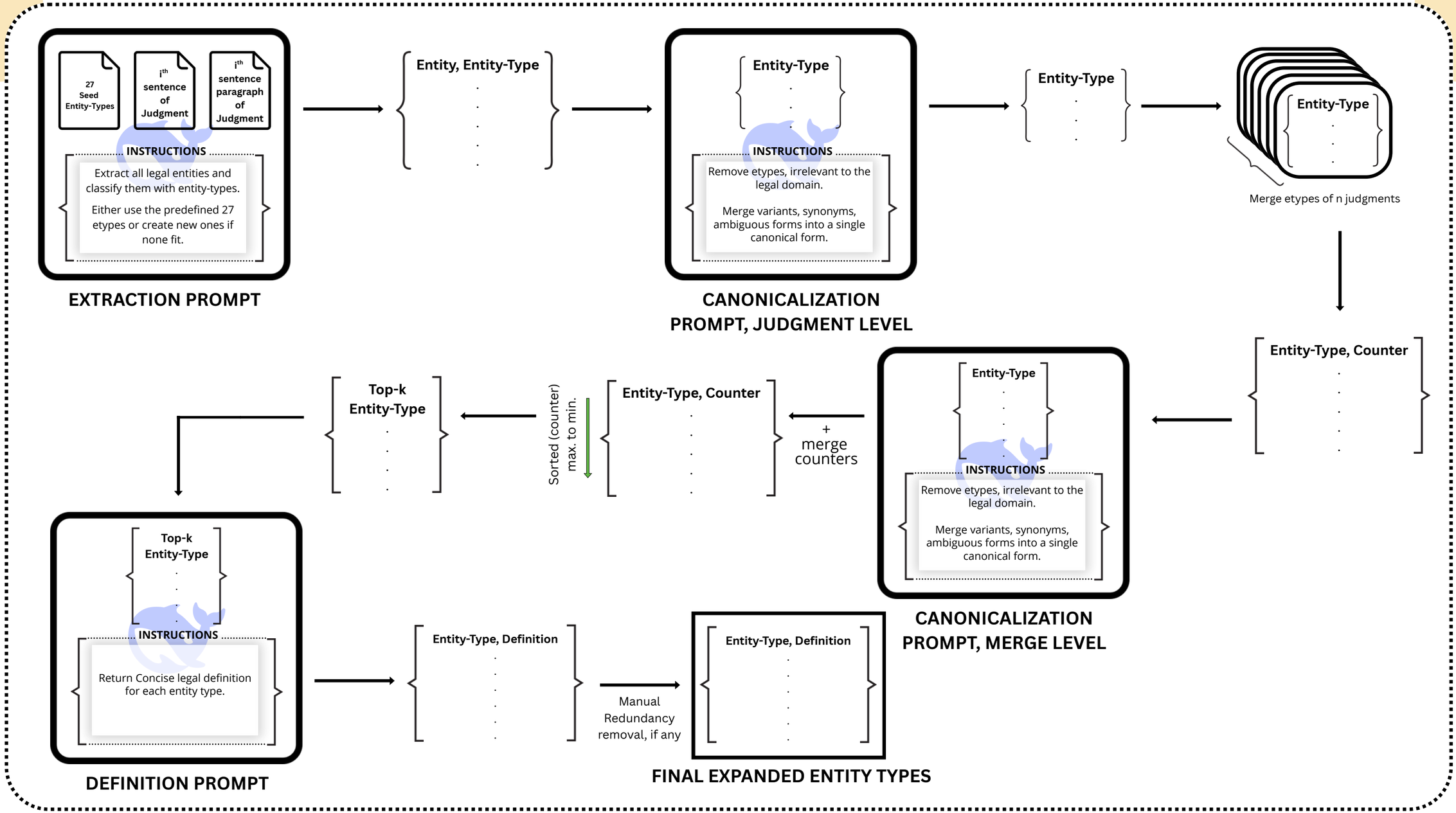
FILOX: Framework for Indian Legal Ontology Expansion

1. **Preprocessing Module**
2. **Entity-Types Expansion Module**
   a. Extraction
   b. Canonicalization (Judgment level)
   c. Canonicalization (Merge level)
   d. Definition Generation
3. **Relations Expansion Module**
   a. Extraction
   b. Canonicalization (Judgment level)
   c. Canonicalization (Merge level)
   d. Definition Generation

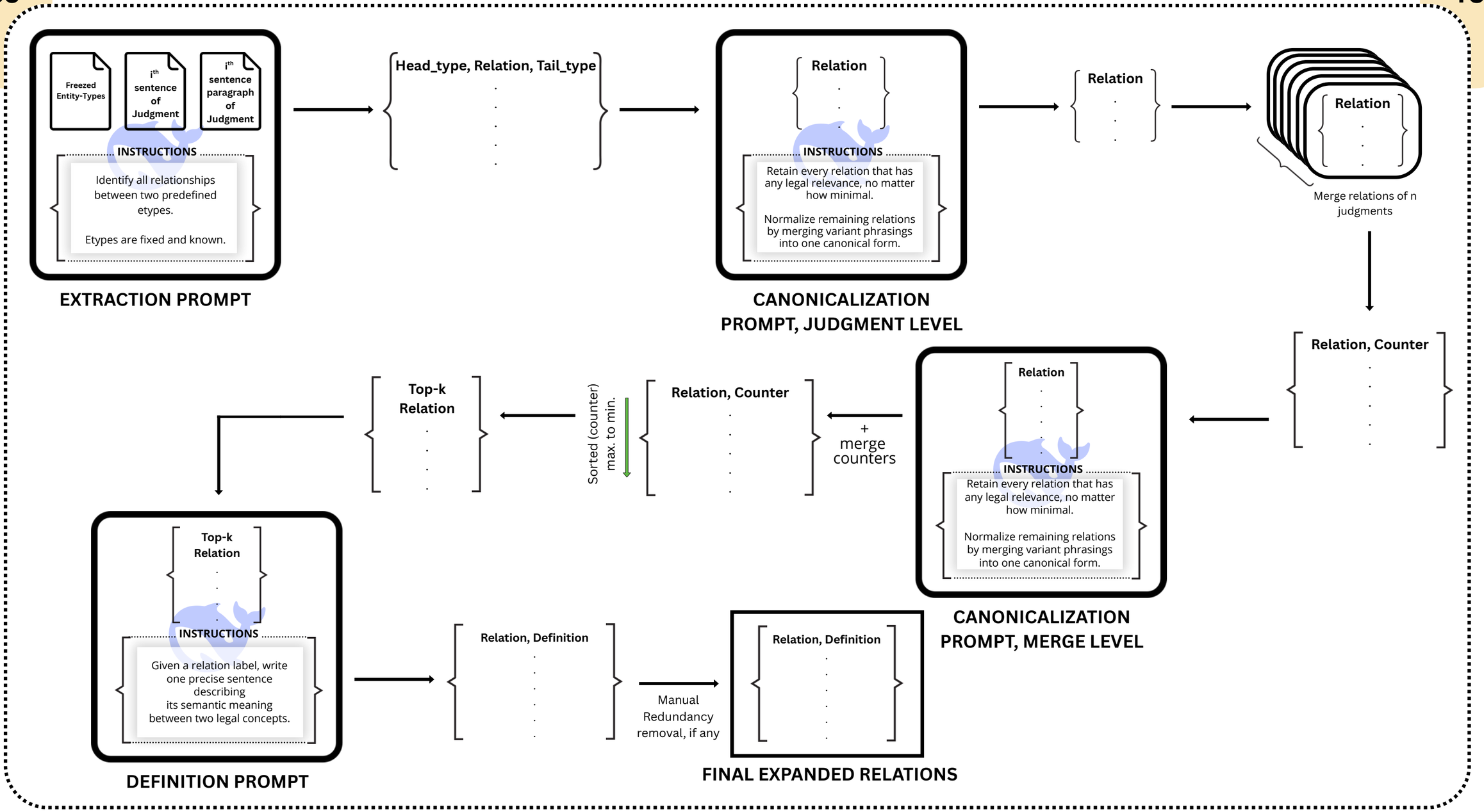
PREPROCESSING MODULE



ENTITY TYPES EXPANSION MODULE



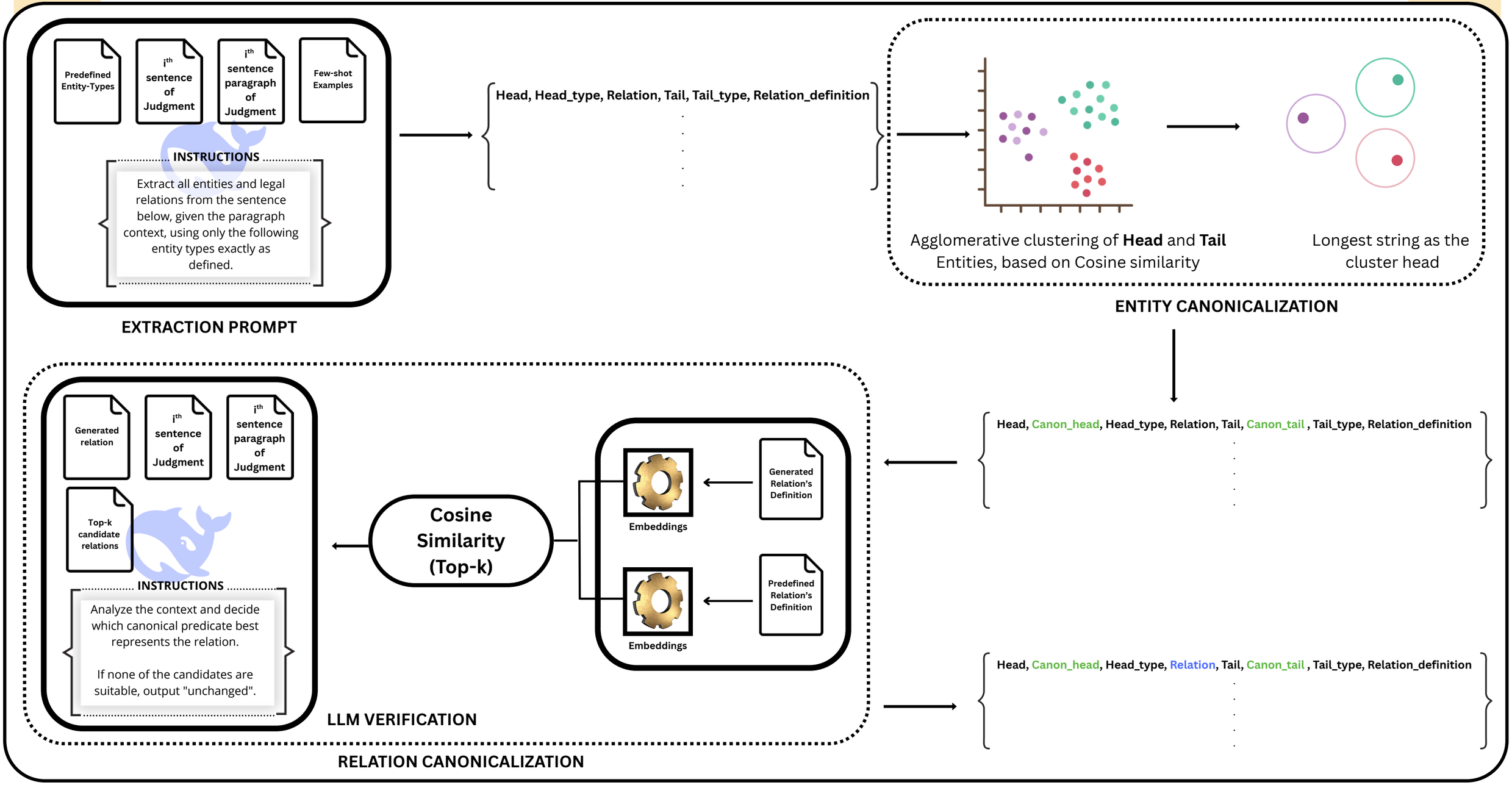
RELATIONS EXPANSION MODULE



FILKGC: Framework for Indian Legal Knowledge Graph Construction

1. **Triplet Extraction**
2. **Entity Canonicalization**
3. **Relation Canonicalization**

Information Extraction for KGC



IMPLEMENTATION AND EXPERIMENTAL RESULTS

Component	Details
RAM	512 GB
Processor	Intel® Xeon® Gold 6326 CPU @ 2.90 GHz (2 processors)
OS Type	64-bit operating system, x64-based processor
OS Name	Windows Server 2022 Standard
GPU	NVIDIA A100
VRAM	80 GB
Python Version	3.12.3

Table 4.1: Hardware Configuration of the server

Metric	Value
Total # judgments	114
Total # words	5,59,034
Total # sentences	18,071
Average # words per judgment	4,903.8
Average # sentences per judgment	158.51

Table 4.2: Statistics of the judgment dataset

IMPLEMENTATION AND EXPERIMENTAL RESULTS

Entity-type Counts Distribution

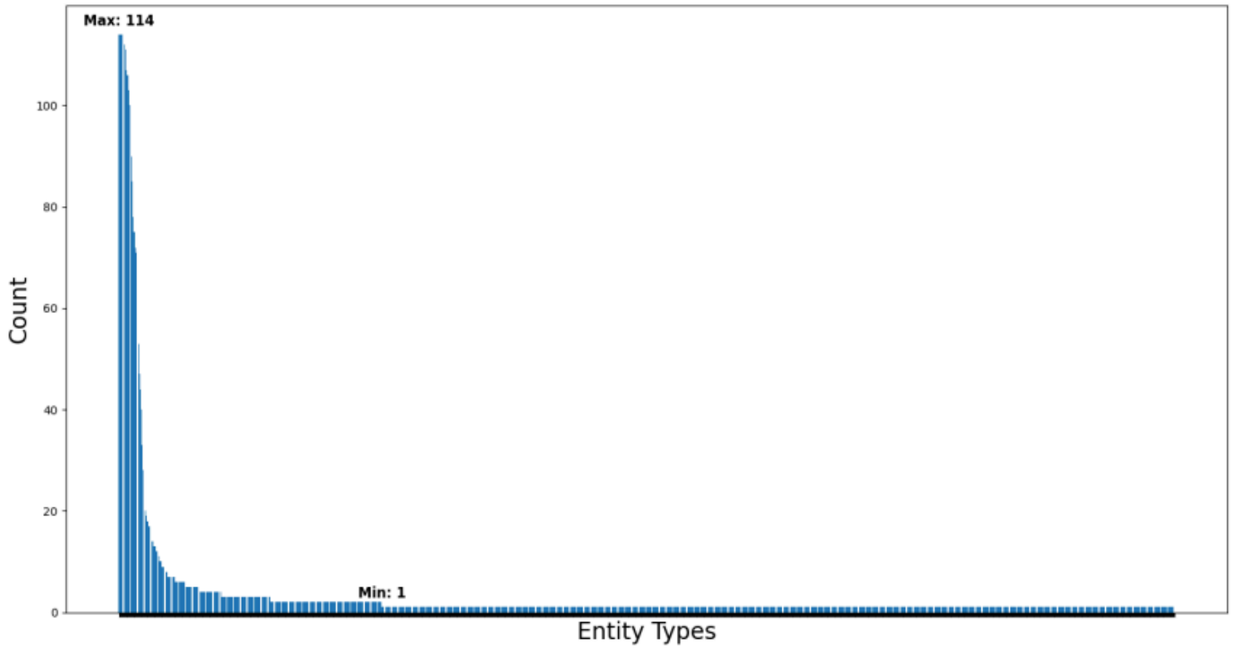


Figure 4.1: Distribution of the count of judgments (y-axis) in which the *entity-types* (x-axis) occur at least once

IMPLEMENTATION AND EXPERIMENTAL RESULTS

Step 1: Extraction; **1427** entity-types

Step 2: Judgment level canonicalization; **674** entity-types

- Date and Time, date amount, Date and Time, date time, DateAndTime, Dates, dates and times, DateTime, Date-Time, etc.- > Date_and_Time
- Removed irrelevant to legal domain such as: Mental Capacity, train, etc.

Step 3: Merge level canonicalization; **587** entity-types

Step 4: Selected top-k: **100** entity-types

Step 5: After manual removal: **87** entity-types

- (legislature body, legislature), (judge, court official judge)

Total Entity-types = 87

S.No	Entity Type	Definition
1	accused	The individual legally charged with committing a crime, against whom prosecution is being pursued.
2	act	A formal written law passed by a legislature.
3	action	A legal claim or cause of action recognized by law that enables an individual or entity to seek redress or remedy through judicial proceedings for an injury or wrong committed.
4	agent	A legal entity designated to act on behalf of another person or organization, having the authority to make decisions or perform actions that legally bind the principal.
5	appeal	A formal request made to a higher court or tribunal to review and overturn a lower court's decision.
6	appellant	A party who brings an appeal, challenging a lower court's decision as incorrect or unlawful.
7	application	A formal request made to a court, tribunal, or other legal authority for a specific action or benefit.
8	argument	A statement or series of statements made to establish or prove a point of law, typically presented during legal proceedings such as trials or hearings.
9	asset	A legally recognized item, property, or right that is owned by an individual, business, or entity and has value in terms of money, benefits, or other resources. It includes tangible items (e.g., real estate, vehicles) and intangible items (e.g., intellectual property, shares).
10	authority	The right or power to make decisions, enforce rules, or take action within a legal framework.
11	beneficiary	A person legally entitled to receive benefits or advantages from another, typically under a trust, will, insurance policy, or other legal arrangement.
12	business	An organization engaged in commercial, industrial, or professional activities, typically aiming to make a profit.
13	case	A legal proceeding in which one party seeks to enforce rights against another party through a court or tribunal.
14	charge	A formal accusation in criminal law alleging that an individual has committed a specific offense.
15	citation	A reference to an authority or source cited as evidence or justification for a statement or proposition.

... etc.

IMPLEMENTATION AND EXPERIMENTAL RESULTS

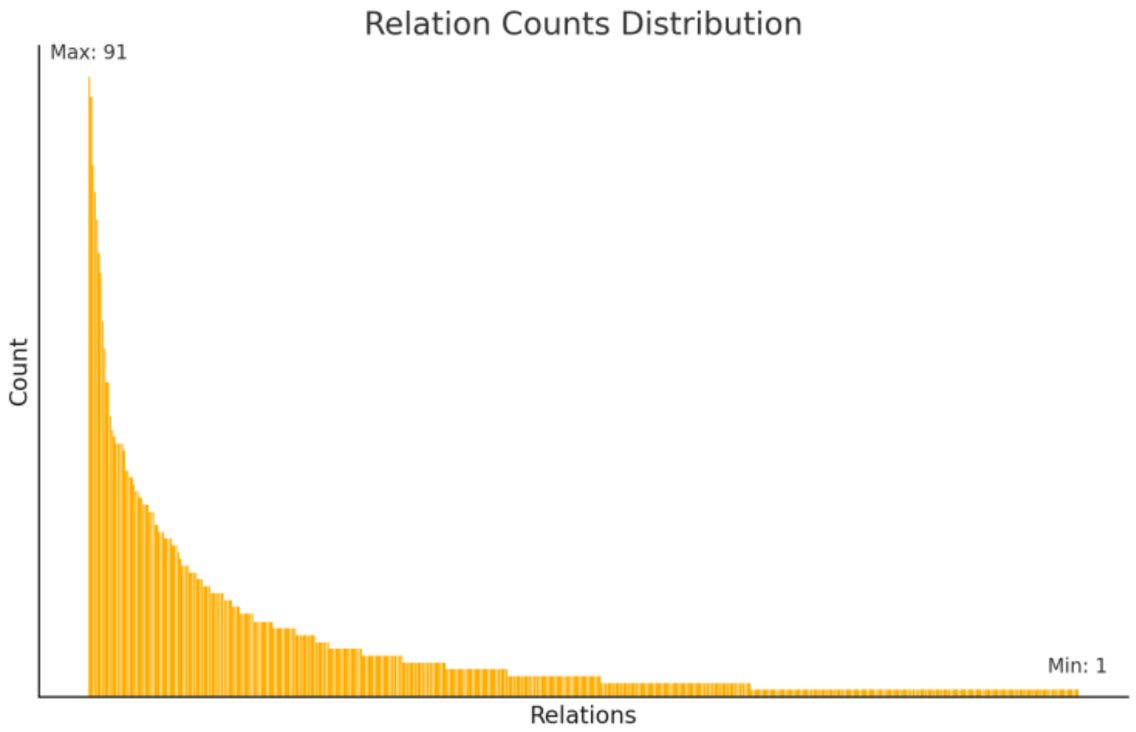


Figure 4.2: Distribution of the count of judgments (y-axis) in which the relations (x-axis) occur at least once

IMPLEMENTATION AND EXPERIMENTAL RESULTS

Step 1: Extraction; **3669** relations

Step 2: Judgment level canonicalization; **1118** relations

- (accused, accused in case, accused of), (testify, testified, testifies, testifies against, testifies in, testifying in), etc.

Step 3: Merge level canonicalization; **602** relations

Step 4: Selected top-k: **320** relations

S.No	Relations	Definition
91	defendant	The party accused of a legal wrongdoing or breach who is required to answer the plaintiff's claims in court.
92	define	One concept defines another when it provides a detailed explanation or specifies the scope and meaning of the latter.
93	delegate	The transfer of authority from one party to another to act on their behalf within specified limits.
94	deliver	To transfer or hand over something, such as property or documents, from one party to another as required by law.
95	demand	A formal request for specific action or compensation.
96	deny	One party refuses to accept or recognize the validity or enforceability of another's claims or rights.
97	depose	To compel a witness to provide sworn testimony in a legal proceeding.
98	deprive	To cause another person to lose a right, entitlement, or possession through wrongful conduct.
99	designation	The formal naming or appointment of a person to a specific role or position under the law.
100	determination	A formal decision made by a legal authority to settle a matter or issue.
101	dictate	One legal concept imposes or requires the existence or implementation of another.
102	did_not_commit	One party is not legally responsible for the actions or obligations of another.
103	direct	A direct relationship exists when one legal concept is immediately connected to another without intermediaries.
104	disallow	Disallow refers to the act where one legal concept restricts or prohibits another from being implemented or enforced.
105	discharge	To release or rid oneself of a legal obligation or liability.

... etc.

ASSIGNING CANONICAL NAMES TO THE ENTITIES

S.No.	Model Name	# Parameters	Details
1	nlpaueb/legal-bert-base-uncased	~110M	Finetuned on European court data
2	nlpaueb/legal-bert-small-uncased	~42M	Finetuned on European court data
3	bhavyagiri/InLegal-Sbert	~110M	Finetuned for Indian legal domain
4	Stern5497/Sbert-legal-xlm-roberta-base	~125M	Finetuned on Legal data
5	Sentence-transformers/all-MiniLM-L6-v2	~22M	Fast, Good, for general semantics

Table 4.7: Choosing embedding models

Comparison file: Link

CHOOSING DOCUMENT CONTEXT WINDOW LENGTHS

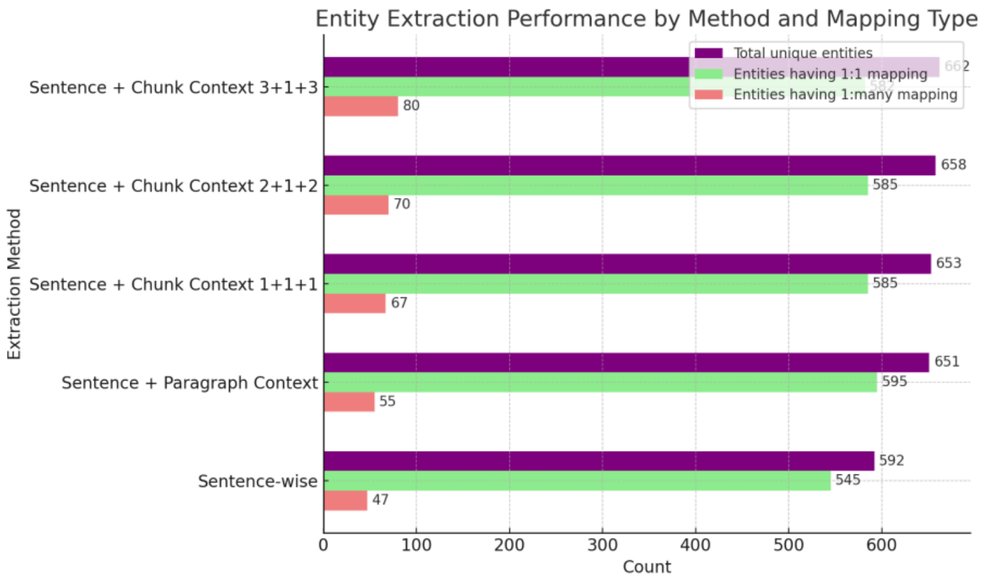


Figure 4.4: Distribution of the five context window lengths (y-axis) against the counts of unique entities, entities with 1:1 mappings, and entities with 1:many mappings (x-axis) **(Before Canonicalization)**

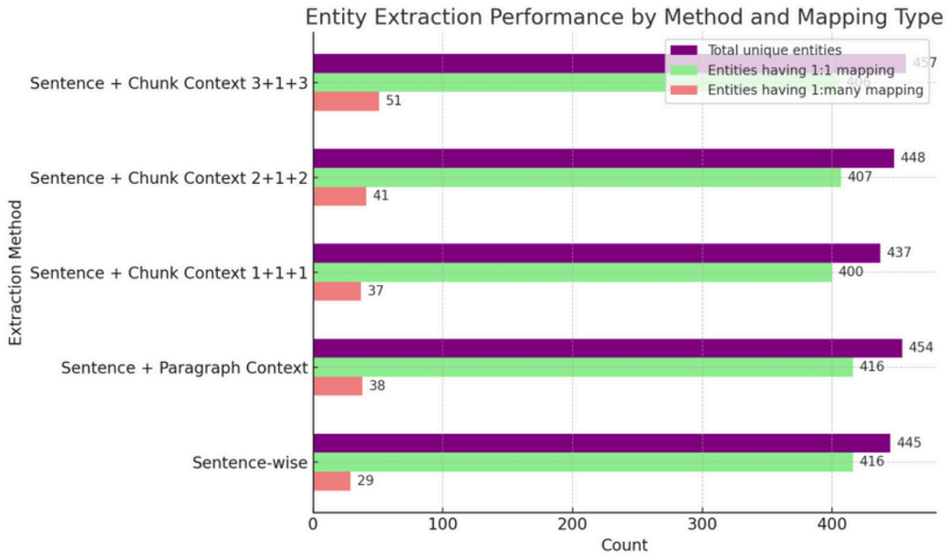


Figure 4.5: Distribution of the five context window lengths (y-axis) against the counts of unique entities, entities with 1:1 mappings, and entities with 1:many mappings (x-axis) **(After Canonicalization)**

Conclusion and Future Work

- Expanded legal ontology to **87 entity types** and **320 relations** using FILOX.
- Extracted a total of **36070** triplets using FILKGC from **114 judgments.**
- In cases where annotated data is available, computational methods still outperforms the LLM-based methods.
- More accurate canonicalization techniques could be devised in the future work.

REFERENCES

1. Banko, M., Cafarella, M.J., Soderland, S., Broadhead, M., Etzioni, O. and Center, T., Open Information Extraction from the Web.
2. Fader, A., Soderland, S. and Etzioni, O., Identifying Relations for Open Information Extraction.
3. Schmitz, M., Soderland, S., Bart, R. and Etzioni, O., 2012, July. Open language learning for information extraction. In Proceedings of the 2012 joint conference on empirical methods in natural language processing and computational natural language learning (pp. 523-534).

REFERENCES

4. Del Corro, L. and Gemulla, R., 2013, May. Clausie: clause-based open information extraction. In Proceedings of the 22nd international conference on World Wide Web (pp. 355-366).
5. Bast, H. and Haussmann, E., 2013, September. Open information extraction via contextual sentence decomposition. In 2013 ieee seventh international conference on semantic computing (pp. 154-159). IEEE.
6. https://spacy.io/api/entityrecognizer/

REFERENCES

33

7.Zhang, B. and Soh, H., 2024, November. Extract, Define, Canonicalize: An LLM-based Framework for Knowledge Graph Construction. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (pp. 9820-9836).

8.Kalamkar, P., Agarwal, A., Tiwari, A., Gupta, S., Karn, S. and Raghavan, V., 2022. Named Entity Recognition in Indian court judgments. NLLP 2022, 2022, pp.184-193.

9.Jain, S., Harde, P., Mihindukulasooriya, N., Gosh, S., Bisht, A. and Dubey, A., 2022, May. Constructing a Knowledge Graph from Indian Legal Domain Corpus. In TEXT2KG/MK@ ESWC (pp. 80-93).

THANK YOU

https://github.com/Harsh-Yadav-02/legalkg llm